



Ahmed Aziz Ben Haj Hmida

Machine Learning Engineer, graduate of the École Polytechnique Fédérale de Lausanne (EPFL), specializing in Computer Vision, Large Language Models, and scalable AI systems.

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Education

EPFL – École Polytechnique Fédérale de Lausanne

M.Sc. in Data Science

GPA: 5.3/6

Relevant Coursework: Machine Learning, Natural Language Processing, Computer Vision, Reinforcement Learning, Data Analysis, Database Systems, Data Science for Business

Lausanne, Switzerland

Sep 2023 – Mar 2026

Technical University of Denmark (DTU)

3rd Year of Bachelor - Exchange

GPA: 5.35/6

Lyngby, Denmark

Aug 2022 – Jul 2023

EPFL – École Polytechnique Fédérale de Lausanne

B.Sc. in Communication Systems

GPA: 5.1/6

Lausanne, Switzerland

Sep 2020 – Jul 2023

Pioneer School of Nabeul

Baccalaureate in Mathematics

Nabeul, Tunisia

Sep 2018 – Jun 2019

Professional Experience

Rollomatic SA

Master's Thesis – Curve-Based CNC Tool Similarity Retrieval - Grade 6/6

Le Landeron, Switzerland

Aug 2025 – Mar 2026

- Engineered an end-to-end similarity search system using geometric representations and deep visual embeddings.
- Built scalable indexing and evaluation pipelines for industrial workflows.
- Optimized for industrial workflows using rigorous testability to ensure seamless R&D integration.
- Improved retrieval accuracy by **11%** and reduced latency by **95–99%**.

Supervisors: Prof. Jean-Philippe Thiran - jean-philippe.thiran@epfl.ch, Emna Maghrebi - e.maghrebi@rollomatic.ch

Rollomatic SA

Machine Learning Internship

Le Landeron, Switzerland

Feb 2025 – Aug 2025

- Developed hybrid retrieval pipelines combining structured parameters and visual embeddings.
- Preprocessed industrial datasets and optimized similarity learning workflows.

Supervisor: Emna Maghrebi - e.maghrebi@rollomatic.ch

Research Projects

Reverse Knowledge Distillation and Model Merging

EPFL LTS4 Lab

Aug 2024 – Feb 2025

- Finetuned compact specialized models and transferred their knowledge into larger complex architectures.
- Applied model merging techniques to preserve general capabilities while improving niche-task performance.

Supervisor: Prof. Pascal Frossard - pascal.frossard@epfl.ch, Ortal Senouf

Discovering promising new startups through evaluating teams' potentials

EPFL ENTC Lab

Sep 2023 – Dec 2023

- Used ML methods to infer speakers' MBTI personalities and emotions from entrepreneurial discussions.
- Built predictive models to identify promising teams to create successful startups.

Supervisor: Prof. Davide Bavato

Technical Skills

Languages: Python, Java, SQL, MATLAB, Scala

ML & AI: NumPy, Pandas, scikit-learn, PyTorch, PySpark

Computer Vision: OpenCV, scikit-image, torchvision, Vision Transformers, CNNs, ResNet

Natural Language Processing: spaCy, HuggingFace, Quantization, LoRA, RLHF, PEFT, SentenceTransformers

Agentic AI: LangChain, LangGraph, LlamaIndex, Vector Search, FAISS, PostgreSQL, Prompt Engineering

DevOps & MLOps: Git, Docker, Kubernetes, FastAPI, MLflow, W&B, DVC, GitHub Actions, GCP, AWS, Streamlit

Visualization: Matplotlib, Seaborn, Plotly, Power BI

Academic Projects

LLM Teaching Assistant for MCQ Answering	Feb 2024 - Jun 2024
• Fine-tuned Qwen1.5 using SFT and DPO with RAG and 4-bit quantization for efficient response generation. • Developed a chatbot User Interface and a robust evaluation pipeline using LLM-as-a-Judge methods and cross-validation to ensure pedagogical accuracy.	
Coin Detector and Counter	Feb 2024 - Jun 2024
• Built segmentation and classification pipelines using CNNs, ViTs, and ResNet50. • Detected and identify coins accurately across diverse backgrounds.	
RL for Motor Skill Transfer	Feb 2024 - Jun 2024
<i>EPFL Alexander Mathis Lab</i> • Trained an expert model on the Myosuite baoding balls task and distilled its motor strategies to novice agents. • Improved learning efficiency and performance of novice agents in complex motor control tasks.	
Machine Learning Operations	Jan 2023 - Mar 2023
• Built scalable ML pipelines using PyTorch, Docker, Kubernetes, MLflow, DVC, AWS, and GCP.	

Certifications

Structuring Machine Learning Projects – DeepLearning.AI	Sep 2023 - Oct 2023
Certificate in Artificial Intelligence – GOMYCODE	Jun 2023 - Sep 2023

Leadership and Volunteering

EPFL Coaching – CMS section Coach	Sep 2023 - Jun 2024
Rotary International – Member	Jul 2018 - Apr 2019
Tunisian Red Crescent – Volunteer	May 2018 - Apr 2019

Languages

Arabic (Native), French (Native/C2), English (C1), German (A2)